Week 1 Practical — Explicit Laplacian Smoothing on ETOPO2 (Python-first)

Geophysics IIIA (Potential Fields & Geothermics)

February 22, 2026

Setup

Ok, let's get started by creating a python virtual environment within VS Code that we can use to install all the necessary packages.

Follow the following steps:

1. Download the codes and data file into a directory for week 1 activities.
2. Open **VS Code** and 'Open Folder' with your week 1 activities.
3. Type 'Ctrl+P' to open the command palette. Type '>' to start a command and start typing 'Python: Create Environment'. The command should appear before you finish typing so you can select it.
4. Select venv as your preferred virtual environment. You should see .venv show up as a directory in your folder.
5. Open a terminal (Ctrl+J to open if needed) and type
   `source .venv/bin/activate`
6. To install the required packages into the virtual environment, type
   `pip install numpy xarray matplotlib jupyterlab`

Overview

The objective of this problem set is to gain familiarity with the python environment, the use of Jupyter notebooks, and explore the use of a laplacian for smoothing an image. Implement the **2-D five-point Laplacian** for *iterative smoothing* by

$$H^{(k+1)} = H^{(k)} - \alpha \nabla^2 H^{(k)},$$

then compare how sharp features blur as smoothing increases. This builds intuition that as observations are made by **moving farther from a source produces a smoother map**.

Data

Use an **ETOPO2** NetCDF (e.g., variable `z`). For speed, subset a regional window (e.g., 120–160°E, 45–5°S).

Tasks

1. **Load & visualize** a regional window; plot elevation with consistent color limits.
2. **Implement the explicit 5-point Laplacian** with unit spacing ($\Delta x = \Delta y = 1$) and Neumann edges (replicate border values). Plot $\nabla^2 h$ (red–blue diverging scale) and describe where curvature is largest.

3. **Iterative smoothing:** For $\alpha \in [0.05, 0.2]$ and steps $K = 5, 10, 20, 40$, compute smoothed maps H_K. Display original + two smoothed maps (same color scale). Discuss which features blur first and why.
4. **Profile comparison:** Extract a profile across a sharp feature. Plot original and smoothed profiles together and estimate a *10–90% edge width* before/after smoothing.
5. **Interpretation (200–300 words):** Explain how curvature (large $|\nabla^2 h|$) relates to features that blur first; argue why increased smoothing mimics being farther from sources; explain why the inverse ("downward continuation") is unstable with noise.

Extension (optional)

On a latitude–longitude grid, east–west spacing shrinks with latitude. For uniform angular steps $\Delta\phi$ (latitude) and $\Delta\lambda$ (longitude),

$$\Delta y \approx R\,\Delta\phi, \qquad \Delta x(\phi) \approx R\cos\phi\,\Delta\lambda,$$

so a physically weighted discrete Laplacian is

$$\nabla^2 h \;\approx\; \frac{h_{i+1,j} - 2h_{i,j} + h_{i-1,j}}{\Delta y^2} \;+\; \frac{h_{i,j+1} - 2h_{i,j} + h_{i,j-1}}{\Delta x(\phi_i)^2}.$$

Implement this variant with per-row $\Delta x(\phi_i)$ and scalar Δy, and re-run smoothing with a smaller α.

Submission

- Figure panel: original + two smoothed maps (e.g., $K = 10, 40$), same color limits.
- Profile plot: original vs smoothed, annotated 10–90% edge widths.
- 200–300 word interpretation.
- Code (notebook/script) with your explicit Laplacian implementation.

Marking rubric (10 points)

- Correct explicit Laplacian & sensible boundary handling (3)
- Smoothing runs & plots with consistent color scales (3)
- Interpretation: curvature, distance-as-smoothing, downward continuation caution, edge-width usage (4)

Notes

Use Neumann (replicate) edges to avoid artificial sinks/sources at borders. If you implement the weighted Laplacian, choose a smaller α due to smaller Δx at higher latitudes.